

The pleasurability of scratching an itch: a psychophysical and topographical assessment

G.A. bin Saif,*† A.D.P. Papoiu,* L. Banari,* F. McGlone,‡ S.G. Kwatra,* Y-H. Chan§ and G. Yosipovitch*¶**

Departments of *Dermatology, ¶Neurobiology & Anatomy, and **Regenerative Medicine, Wake Forest School of Medicine, Winston-Salem, NC 27157, U.S.A.

†Department of Dermatology, College of Medicine, King Saud University, Riyadh, Saudi Arabia

‡School of Natural Sciences and Psychology, Liverpool John Moores University, Liverpool, U.K.

§Biostatistics Unit, Yong Loo Lin School of Medicine, National University Health System, Singapore

Summary

Correspondence

Gil Yosipovitch.

E-mail: gyosipov@wakehealth.edu

Accepted for publication

4 January 2012

Funding sources

This work was supported by NIAMS (NIH) grant 5AR0155902-3.

Conflicts of interest

None declared.

G.A.b.S. and A.D.P.P. contributed equally to this report and should be considered first authors.

DOI 10.1111/j.1365-2133.2012.10826.x

Background Scratching an itch is perceived as being pleasurable. However, an analysis of topographical variations in itch intensity, the effectiveness of scratching to provide itch relief and the associated pleasurability has not been performed at different body sites.

Objectives To examine the role of scratching pleasurability in providing itch relief by investigating whether itch intensity is perceived differently at three different sites and to assess a potential correlation between the pleasurability and itch attenuation induced by scratching.

Methods Itch was induced on the forearm, ankle and back using cowhage spicules in 18 healthy subjects. These sites were subsequently scratched by an investigator with a cytology brush immediately following itch induction. The intensity of itch with and without scratching at these sites and the pleasurability of scratching were recorded by taking visual analogue scale ratings at 30-s intervals.

Results Mean itch intensity and scratching pleasurability ratings at the ankle and back were significantly higher than on the forearm. For the forearm and ankle, the greater the itch while scratching, the higher was the pleasurability. A higher baseline itch was linked to a higher itch reduction secondary to scratching in all tested areas. Pleasurability paralleled the curve of itch reduction for the back and forearm; however, scratching pleasurability at the ankle remained elevated and only slightly decreased while itch was diminishing.

Conclusions There are topographical differences in itch intensity, the effectiveness of scratching in relieving itch and the associated pleasurability. Experimental itch induced by cowhage was more intensely perceived at the ankle, while scratching attenuated itch most effectively on the back.

Scratching is not only an effective means of relieving an itch, it is also often perceived as a highly pleasurable experience. The urge to scratch is particularly well known to be hard to resist and is the driving force underlying the formation of the itch-scratch cycle, an addictive and vicious cycle. We have recently reported that ratings of scratching pleasurability in patients with atopic eczema are correlated to itch intensity ratings.¹

Currently, there is a paucity of psychophysical studies examining the pleasurability of scratching. In previous psychophysical studies examining itch, the forearm has been used predominantly as the typical site for itch induction, but much less is known about areas such as the back or the ankle, areas where patients with diseases such as lichen simplex chronicus or atopic eczema frequently itch and scratch. Indeed, the back

is well known as a preferred site for scratching, as proven by the multitude of back scratchers ubiquitous in many cultures for several centuries.

The goal of the current study was to examine the role of the pleasurability of scratching in providing relief for itch. We first evaluated whether itch intensity was perceived differently at different body sites and then we investigated the potential correlation between the pleasurability and the itch relief induced by scratching.

We used a visual analogue scale (VAS) to assess itch intensity, and the pleasurability associated with scratching at forearm, ankle and back. Cowhage, rather than histamine, was selected as the pruritogen to induce experimental itch. Recent studies have suggested that cowhage-induced itch is a more

reliable model for the experimental study of chronic itch^{2,3} as cowhage acts on PAR2 receptors involved in the mediation of itch in atopic eczema and possibly in other dermatoses.

Materials and methods

Design overview

Itch was induced on the volar aspect of the forearm, the anterior side of the ankle region and on the back, approximately 5 cm laterally from the midline, under the scapula. In the next phase, the same sites were scratched with a cytology brush immediately following itch induction. The intensity of itch with and without scratching at these sites and the pleasurability of scratching were assessed. Experiments performed at different body sites were separated by a 10-min interval.

Subjects

Eighteen healthy subjects (10 women, eight men) aged 22–59 years (mean $\pm$ SD 33.5 $\pm$ 10.5) participated in this study. All procedures were approved by the Institutional Review Board of Wake Forest University School of Medicine (WFU Health Sciences). All volunteers provided written informed consent and were free to withdraw from the study at any time.

Itch induction

Itch was evoked by the application of 40–45 cowhage spicules, which were counted under a magnifying lens and subsequently rubbed into the subject's skin by a study investigator. The spicules were rubbed gently in a circular motion for 45 s within a 4 cm² area of skin. Upon itch induction, the spicules were subsequently removed using adhesive tape (Scotch/3M, St Paul, MN, U.S.A.).

Psychophysical assessment of itch intensity and scratching pleasurability

Itch intensity (with and without scratching) and scratching pleasurability were assessed every 30 s for a duration of 5 min using a numerical VAS that was labelled from 0 (no itch) to 10 (maximum unbearable itch).

Passive scratching

Scratching was accomplished by a study investigator continuously moving a cytology brush in a linear direction (Medi-Pak 7-inch cytology brush 24–2199; General Medical Corporation, Elkridge, MD, U.S.A.) over the itchy areas. Uniformity was controlled by applying sufficient pressure to bend skin-facing brush bristles, so that the brush handle touched the skin surface. The bending force of the cytology brush was equivalent to approximately 29 g. The cytology brush includes approximately 1000 individual bristles and the diameter of the

brush is 7.5 mm. Further, the same member of the research team applied the cytology brush for all subjects. Each examinee underwent a training session in which study personnel applied the cytology brush. The examinees were asked whether the procedure was similar to their experience when they scratch their skin themselves. All subjects reported that they perceived this procedure as closely simulating scratching. If any of the examinees perceived application of the cytology brush as painful, scratching was ceased.

Statistical analysis

All statistical analyses were performed using PASW 18.0 software (SAS, Chicago, IL, U.S.A.) with statistical significance set at $P < 0.05$. Paired *t*-tests were performed to compare mean and peak VAS ratings of baseline itch, itch while scratching, and the pleasurability at the three sites (forearm, back and ankle). The results were Bonferroni corrected. A Spearman correlation analysis was performed between ratings of itch intensity, the amplitude of itch reduction and the pleasurability of scratching.

Results

Regional variability in itch perception

Mean and peak itch intensity at the forearm, ankle and back are shown in Table 1 and Figure 1a. A comparison of cowhage-induced itch at these three sites reveals that the mean itch intensity ratings at the ankle and back were significantly higher than at the forearm ($P = 0.002$). No significant difference in itch intensity was found between the ankle and the back.

Pleasurability of scratching

Significant differences in scratching pleasurability were noted among the different areas: the mean scratching pleasurability ratings for the ankle and back were significantly higher than for the forearm, as shown in Figure 1c ($P = 0.02$ and $P = 0.007$, respectively). There was no significant difference

Table 1 Mean and peak cowhage itch intensity reported at the three areas tested

	Itch intensity		
	Ankle	Back	Forearm
In the absence of scratching			
Mean $\pm$ SD	6.0 $\pm$ 2.2	5.8 $\pm$ 2.4	3.8 $\pm$ 2.2
Peak, mean $\pm$ SD	8.3 $\pm$ 2.0	8.6 $\pm$ 1.3	6.7 $\pm$ 2.3
While scratching			
Mean $\pm$ SD	2.5 $\pm$ 1.5	2.7 $\pm$ 1.95	1.9 $\pm$ 1.8
Peak, mean $\pm$ SD	6.2 $\pm$ 2.1	5.4 $\pm$ 2.5	4.3 $\pm$ 3.1

Values were measured on a numerical visual analogue scale of 0–10 (0, no itch; 10, maximum unbearable itch) before and during scratching.

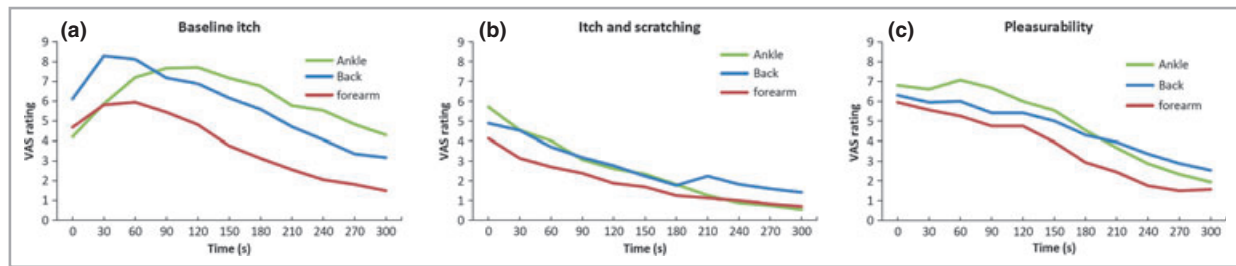


Fig 1. Mean itch intensity induced by cowhage (a) in the absence of scratching and (b) during scratching, and (c) the pleasurability ratings of scratching, recorded simultaneously with (b), at three sites: forearm, ankle and back. Scratching reduced itch intensity and was paralleled by the pleasurability (ratings) curve. VAS, visual analogue scale.

in scratching pleasurability between the ankle and the back (Table 2). Age and sex differences had no effect on itch perception, scratching-induced itch modulation, or scratching pleasurability.

The higher the itch ratings while scratching, the higher was the mean pleasurability rating at the forearm and ankle ($P = 0.038$ and $P = 0.047$; Spearman's rank correlation coefficient 0.523 and 0.52, respectively).

Scratching reduces itch intensity

Mean itch intensity ratings were significantly reduced by scratching in comparison with control itch ratings with no scratching in all three areas tested as shown in Figure 1b ($P = 0.001$, $P < 0.001$ and $P = 0.001$ at the ankle, back and forearm, respectively). The peak intensities of itch were also significantly reduced by scratching in all areas tested ($P = 0.001$, $P < 0.001$ and $P < 0.001$ at the ankle, back and forearm, respectively). Peak and mean itch intensities while scratching are shown in Table 1.

A comparison of mean itch intensities while scratching among the tested areas revealed that itch intensity on the back was significantly higher than on the forearm ($P = 0.009$). No significant differences were noted between the ankle and back or ankle and forearm.

The higher the mean baseline itch rating, the higher was the itch reduction secondary to scratching in all tested areas ($P = 0.042$, $P < 0.001$, $P = 0.001$; Spearman's rank correlation coefficient 0.513, 0.814 and 0.746 for the forearm, ankle and back, respectively). The mean $\pm$ SD itch reduction was 3.59 ± 2.0 , 3.05 ± 2.6 and 1.9 ± 1.8 at the ankle, back and forearm, respectively (Fig. 2). Comparing the itch reduction in the three areas showed that itch relief at the ankle and

back was significantly higher than on the forearm ($P = 0.005$ and $P = 0.038$, respectively).

When itch reduction is plotted in the same graph with scratching pleasurability, a trend becomes apparent: ratings of pleasurability clearly parallel the curve of itch reduction for the back and forearm. In contrast, scratching pleasurability remains elevated at the ankle and only slightly decreases while itch is relieved (Fig. 3).

Discussion

The results of this study show topographical differences in itch intensity, itch attenuation by scratching, and the associated pleasurability. Cowhage-induced itch was perceived most intensely at the ankle, while the attenuation of itch by scratching was most effective on the back. In both measurements, the intensity of perceived itch, as well as the magnitude of itch relief induced by scratching, were less pronounced on the forearm compared with the other two sites.

Similarly, pleasurability reached higher amplitudes at the ankle, followed by the back and forearm. Interestingly, scratching on the back and forearm was found to attenuate cowhage-induced itch in a parallel fashion to the scratching pleasurability curve: the greater the itch ratings, the greater was the pleasurability of scratching. However, the pleasurability of scratching the ankle appears to be longer lived compared with the other two sites.

While we previously reported that pleasurability correlates with itch intensity in patients with atopic eczema,¹ topographical differences in perceived itch intensity have seldom been reported. Topographical variations in itch perception have previously been reported by Shelley and Arthur, who noted remarkable differences in itch perception among various body areas in response to the insertion of a single cowhage spicule.⁴ In addition, Truini *et al.* found that the intensity of histamine-induced itch increased from face to foot, in the opposite direction to the topographical variations in warmth and burning sensations.⁵ We could not directly compare these findings with ours, as we examined different body areas and, moreover, we used cowhage as an itch inducer (cowhage releases mucunain which works via PAR2 receptors). Many factors could be envisioned to play a role in the topographical varia-

Table 2 Mean and peak scratching pleasurability as measured by a visual analogue scale (0–10) during cowhage-induced itching in the three areas tested

	Ankle	Back	Forearm
Mean $\pm$ SD	4.91 $\pm$ 2.81	4.64 $\pm$ 2.73	3.67 $\pm$ 2.51
Peak, mean $\pm$ SD	7.87 $\pm$ 1.92	7.06 $\pm$ 2.75	6.56 $\pm$ 3.08

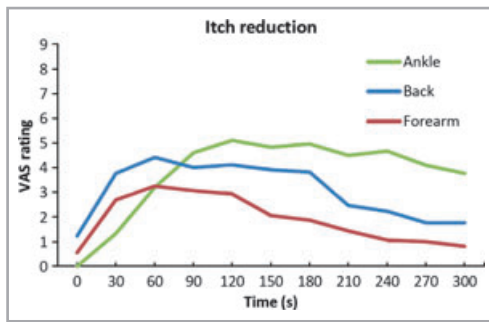


Fig 2. Mean itch reduction at the forearm, ankle and back due to scratching a cowhage-induced itch. The highest itch reduction was at the ankle, followed by the back and the forearm. VAS, visual analogue scale.

tions observed. The first of these is nerve density. Skin biopsy studies have shown a regional variability in density of epidermal free nerve endings, which are higher in proximal, as opposed to distal (leg) sites.^{6–8} However, these findings do not explain why the ankle, which is the most distal of the sites examined, was the itchiest site. Interestingly, we also have previously shown nerve fibre dropout to be associated with lichen amyloidosis and nummular eczema, two chronic itchy dermatoses often located in the ankle (Maddison *et al.*).^{9,10}

In addition, spatial summation of itch information related to variations in the density of receptors may underlie differences in itch sensitivity. Spatial effects based on nociceptor density have been proposed for the sensation of warmth, cold, heat pain and cold pain.^{11–15} There are also marked regional variations in the level of several neuropeptides that play a role in itch induction (substance P, calcitonin gene-related peptide).¹⁶ Regional or topographical differences in whole-body distribution of endocannabinoid and/or opioid receptors could also underlie regional differences in itch sensitivity and pleasurability of scratching.

It is also important to highlight the larger context of our findings in comparing differences between itch and pain transmission. While pain sensation is associated with a withdrawal (reflex), itch leads to a scratching response that is, moreover,

pleasurable. However, scratching could also induce an algedonic response (characterized by the association of pleasure and pain) that could result in the inhibition of itch, according to the gate theory.¹⁷ Overall, itch seems to encompass more hedonic aspects that may vary topographically. An explanation for this contrast may lie in differences in central neuronal processing.

Itch relief induced by scratching is postulated to work via central nervous system suppression of itch processing.¹⁸ Andrew and Craig identified a population of spinothalamic tract (STT) neurones that responded to itch inducers.¹⁹ Davidson *et al.* have more recently shown that in primates, scratching the skin within the receptive field of STT neurones inhibited their firing rates during histamine-evoked activity.¹⁸ This mechanism suggests that scratching attenuates the spinothalamic transmission of itch, and therefore its subsequent perceptual and affective qualities. Scratching also activates brain areas that process itch sensation, while deactivating areas in the brain that are involved in the unpleasant sensation of itch, such as the anterior cingulate cortex.²⁰ Neuroimaging studies in healthy volunteers have shown that passive (artificial) scratching performed by an investigator, as well as the sensation of itch itself, evoke a strong activation of the putamen, which is not seen during passive scratching in the absence of itch.²¹ The putamen is a part of the dorsal striatum, which is involved in the anticipation of pleasure.²² Of particular relevance to scratching behaviour, Robbins and Everitt showed that dopamine in the dorsal striatum plays a role in the sensorimotor coordination of goal-directed consummatory behaviours and the initiation of an appropriate 'response set', such as the motor reparatory processes required for scratching.²³

The peripheral nervous system may also play a powerful role in the relief of itch by scratching. Itch sensation is well known to be mediated by a subpopulation of cutaneous afferent C-fibres. Recent evidence indicates that the density of cutaneous C-fibres is linked to the perceived pleasantness of touch.^{24,25} This latter observation builds on evidence that tactile pleasure is suggested to be mediated by a newly characterized mechanosensitive cutaneous afferent C-fibre system – C-tactile afferents.²⁶ These factors may provide evidence for a

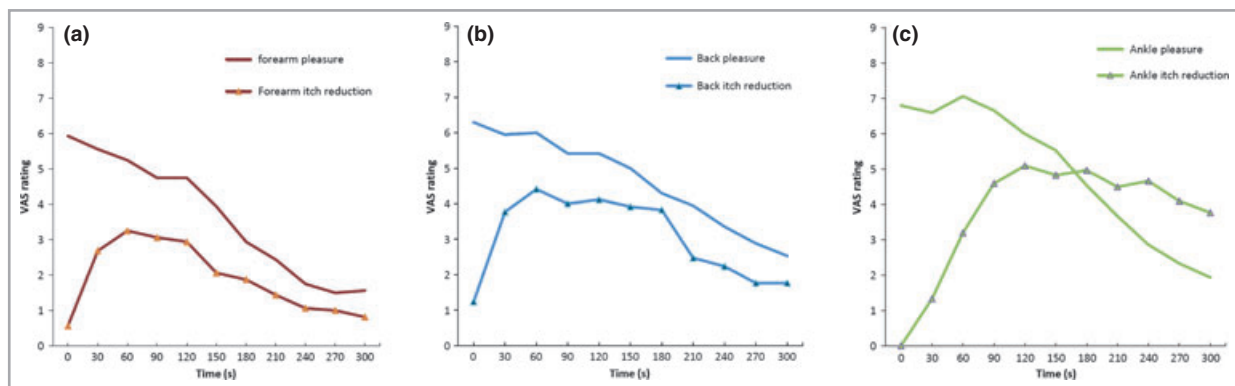


Fig 3. Itch reduction induced by scratching and scratching pleasurability plots over time follow a similar descending curve for forearm and back (a, b) whereas pleasurability remains slightly elevated while itch is relieved on the ankle (c). VAS, visual analogue scale.

peripheral nerve mechanism underpinning the reward basis of scratching itchy skin, in which C-tactile afferents provide the reward signals driving scratching behaviour. It has also been suggested that a better-known afferent C-fibre system – the C-nociceptors – inhibits itch neurones in the dorsal horn of the spinal cord. Another possible mechanism for itch suppression by scratching could be related to a release of prostaglandin D₂. Prostaglandin D₂ was reported to play an inhibitory role against pruritus in a rodent model of atopic eczema,⁶ but it has not been studied in relation to pleurability and the applicability to humans remains to be proven.

The present study uncovers a topographical relationship between itch attenuation by scratching and the accompanying pleurability in healthy individuals. Future studies could also examine the scratching pleurability associated with other itchy areas such as the scalp²⁷ or the anogenital region. It would also be clinically valuable to examine further topographical differences in chronic itch diseases where itch exhibits site predilections, such as atopic eczema, lichen simplex chronicus or lichen amyloidosis.

What's already known about this topic?

- The forearm has been used predominantly as the typical model for itch induction, but less is known about the back or the ankle, areas where patients with lichen simplex chronicus or atopic eczema frequently itch and scratch.
- The pleurability associated with scratching an itch at different sites has not been previously reported.

What does this study add?

- We evaluated topographical differences in perception of itch induced with cowhage at three different body sites: forearm, ankle and back.
- We also investigated the effectiveness of scratching in providing itch relief at these sites in relationship with the accompanying pleurability.

References

- O'Neill JL, Chan YH, Rapp SR, Yosipovitch G. Differences in itch characteristics between psoriasis and atopic dermatitis patients: results of a web-based questionnaire. *Acta Derm Venereol* (Stockh) 2011; **91**:537–40.
- Papoiu AD, Tey HL, Coghill RC *et al.* Cowhage-induced itch as an experimental model of pruritus. A comparison with histamine-induced itch. *PLoS ONE* 2011; **6**:e17786.
- Kosteletzky F, Namer B, Forster C, Handwerker HO. Impact of scratching on itch and sympathetic reflexes induced by cowhage (*Mucuna pruriens*) and histamine. *Acta Derm Venereol* (Stockh) 2009; **89**:271–7.
- Shelley W, Arthur R. The neurohistology and neurophysiology of the itch sensation in man. *Arch Dermatol* 1957; **76**:296–323.
- Truini A, Leone C, Di Stefano G *et al.* Topographical distribution of warmth, burning and itch sensations in healthy humans. *Neurosci Lett* 2011; **494**:165–8.
- Sugimoto M, Arai I, Futaki N *et al.* Putative mechanism of the itch-scratch circle: repeated scratching decreases the cutaneous level of prostaglandin D₂, a mediator that inhibits itching. *Prostaglandins Leukot Essent Fatty Acids* 2007; **76**:93–101.
- Lauria G, Holland N, Hauer P *et al.* Epidermal innervation: changes with aging, topographic location, and in sensory neuropathy. *J Neurol Sci* 1999; **164**:172–8.
- McArthur JC, Stocks EA, Hauer P *et al.* Epidermal nerve fiber density: normative reference range and diagnostic efficiency. *Arch Neurol* 1998; **55**:1513–20.
- Maddison B, Namazi MR, Samuel LS *et al.* Unexpected diminished innervation of epidermis and dermoepidermal junction in lichen amyloidosis. *Br J Dermatol* 2008; **159**:403–6.
- Maddison B, Parsons A, Sanguenza O *et al.* Retrospective study of intraepidermal nerve fiber distribution in biopsies of patients with nummular eczema. *Am J Dermatopathol* 2011; **33**:621–3.
- Essick G, Guest S, Martinez E *et al.* Site-dependent and subject-related variations in perioral thermal sensitivity. *Somatosens Mot Res* 2004; **21**:159–75.
- Stevens JC, Marks LE. Spatial summation of cold. *Physiol Behav* 1979; **22**:541–7.
- Kojo I, Pertovaara A. The effects of stimulus area and adaptation temperature on warm and heat pain thresholds in man. *Int J Neurosci* 1987; **32**:875–80.
- Douglass DK, Carstens E, Watkins LR. Spatial summation in human pain perception. *Soc Neurosci Abstr* 1988; **18**:710.
- Douglass DK, Carstens E, Watkins LR. Spatial summation in human thermal pain perception: comparison within and between dermatomes. *Pain* 1992; **50**:197–202.
- Eedy DJ, Shaw C, Johnston CF, Buchanan KD. The regional distribution of neuropeptides in human skin as assessed by radioimmunoassay and high-performance liquid chromatography. *Clin Exp Dermatol* 1994; **19**:463–72.
- Melzack R, Wall PD. Pain mechanisms: a new theory. *Science* 1965; **150**:971–9.
- Davidson S, Zhang X, Khasabov SG *et al.* Relief of itch by scratching: state dependent inhibition of primate spinothalamic tract neurons. *Nat Neurosci* 2009; **12**:544–6.
- Andrew D, Craig AD. Spinothalamic lamina I neurons selectively sensitive to histamine: a central neural pathway for itch. *Nat Neurosci* 2001; **4**:72–7.
- Yosipovitch G, Ishiuchi Y, Patel TS *et al.* The brain processing of scratching. *J Invest Dermatol* 2008; **128**:1806–11.
- Vierow V, Fukuoka M, Ikoma A *et al.* Cerebral representation of the relief of itch by scratching. *J Neurophysiol* 2009; **102**:3216–24.
- McLean J, Brennan D, Wyper D *et al.* Localization of regions of intense pleasure response evoked by soccer goals. *Psychiatry Res* 2009; **171**:33–43.
- Robbins T, Everitt B. Functions of dopamine in the dorsal and ventral striatum. *Semin Neurosci* 1992; **4**:119–27.
- Morrison I, Löken LS, Minde J *et al.* Reduced C-afferent fibre density affects perceived pleasantness and empathy for touch. *Brain* 2011; **134**:1116–26.
- Löken LS, Wessberg J, Morrison I *et al.* Coding of pleasant touch by unmyelinated afferents in humans. *Nat Neurosci* 2009; **12**:547–8.
- McGlone F, Spence C. The cutaneous senses: touch, temperature, pain/itch, and pleasure. *Neurosci Biobehav Rev* 2010; **34**:145–7.
- bin Saif GA, Ericson ME, Yosipovitch G. The itchy scalp – scratching for an explanation. *Exp Dermatol* 2011; **20**:959–68.